

ELITE MATHEMATICS

Pearson Edexcel International A Level

Pure 1

Adaptive Classified

Chapter 02 · Quadratics

With Answers

Questions with full worked solutions

- 42 pages
- Past papers up to June 2026

Quadratics

Your classified route

15
questions

105
marks

5
families

10
variants

DR ESLAM AHMED

CLASSIFIED BY IDEA

Move from recognition to method, then use the working space to complete the solution.

Quadratics

Families and variants

F01 • Quadratic Equation Methods

V01 Direct Exact Quadratic Solution

V02 Square-root or Fractional-power Substitution

V03 Even-power or Biquadratic Substitution

V04 Odd-power Quadratic Substitution

F02 • Completed-square Form and Parabola Features

V01 Complete the Square, Read and Sketch

V02 Complete the Square, then Intersect and Define a Region

F03 • Construct a Quadratic from Graph Data

V01 Roots, Symmetry and a Point

V02 Vertex Form and a Point

F04 • Discriminant and Root-count Conditions

V01 Parameter Range from a Required Root Count

V02 Curve Tangency or Separation

F05 • Quadratic Modelling from Constraints

V01 Sum-product Geometric Constraints

Quadratic Equation Methods

WHAT THIS FAMILY ASKS

Solve a quadratic directly or reveal one by substituting for a repeated expression, then map all admissible roots back to the original variable.

VARIANT ROUTE

V02 Square-root or Fractional-power Substitution

V03 Even-power or Biquadratic Substitution

3 Complete Classified questions

19 total marks

8. Solve, using algebra, the equation

$$x - 6x^{\frac{1}{2}} + 4 = 0$$

Fully simplify your answers, writing them in the form $a + b\sqrt{c}$, where a , b and c are integers to be found.

(5)

(Total 5 marks)

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Question 1

WMA11/01 October 2019

T02 Quadratics

F01 Quadratic Equation Methods

V02 Square-root
Fractional-power
or

Substitution

1

Use a substitution

Let $u = \sqrt{x}$. Then $x = u^2$, so the equation becomes $u^2 - 6u + 4 = 0$.

2

Solve the quadratic

$$u = \frac{6 \pm \sqrt{36 - 16}}{2} = \frac{6 \pm 2\sqrt{5}}{2} = 3 \pm \sqrt{5}.$$

3

Return to x

Since $u = \sqrt{x}$, $x = u^2 = (3 \pm \sqrt{5})^2 = 9 + 5 \pm 6\sqrt{5} = 14 \pm 6\sqrt{5}$.

FINAL ANSWER

$$x = 14 + 6\sqrt{5} \text{ or } x = 14 - 6\sqrt{5}.$$

5. (a) Find, using algebra, all solutions of

$$20x^3 - 50x^2 - 30x = 0 \quad (3)$$

- (b) Hence find all real solutions of

$$20(y + 3)^{\frac{3}{2}} - 50(y + 3) - 30(y + 3)^{\frac{1}{2}} = 0 \quad (4)$$

(Total 7 marks)

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Question 2

WMA11/01 January 2020

T02 Quadratics

F01 Quadratic Equation Methods

V02 Square-root
Fractional-power
or

Substitution

1

Factorise the cubic

$$20x^3 - 50x^2 - 30x = 10x(2x^2 - 5x - 3) = 10x(2x + 1)(x - 3).$$

2

Solve part (a)

$$10x(2x + 1)(x - 3) = 0, \text{ so } x = 0, x = -\frac{1}{2}, \text{ or } x = 3.$$

3

Use the same structure for part (b)

Let $u = (y + 3)^{1/2}$. Then the equation becomes $20u^3 - 50u^2 - 30u = 0$, so $u = 0, u = -\frac{1}{2}, \text{ or } u = 3$.

4

Reject the invalid square-root value

Since $u = \sqrt{y + 3} \geq 0$, reject $u = -\frac{1}{2}$. If $u = 0$, then $y = -3$. If $u = 3$, then $y + 3 = 9$, so $y = 6$.

FINAL ANSWER

$$(a) x = 0, -\frac{1}{2}, 3. \quad (b) y = -3, 6.$$

4.

**In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

(i) Using the laws of indices, solve

$$3^{2p-5} = \frac{1}{9}\sqrt{27}$$

(3)

(ii) Find the real roots of the equation

$$x^2 + 1 = \frac{36}{x^2 - 4}$$

(Total for Question 4 is 7 marks)

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Question 3

WMA11/01 October 2025

T02 Quadratics

F01 Quadratic Equation Methods

V03 Even-power or Bi-quadratic Substitution

1

Rewrite the right-hand side

$$\frac{1}{9}\sqrt{27} = 3^{-2} \cdot 3^{3/2} = 3^{-1/2}.$$

2

Equate powers of 3

$$3^{2p-5} = 3^{-1/2}, \text{ so } 2p - 5 = -\frac{1}{2}.$$

3

Solve for p

$$2p = \frac{9}{2}, \text{ hence } p = \frac{9}{4}.$$

4

Clear the denominator

$$x^2 + 1 = \frac{36}{x^2 - 4} \text{ gives } (x^2 + 1)(x^2 - 4) = 36.$$

5

Form a quadratic in x squared

$$x^4 - 3x^2 - 4 = 36, \text{ so } x^4 - 3x^2 - 40 = 0.$$

6

Factor

$$(x^2 - 8)(x^2 + 5) = 0.$$

7

Use real roots

$x^2 = 8$, so $x = \pm 2\sqrt{2}$. The equation $x^2 = -5$ gives no real roots.

FINAL ANSWER

(i) $p = \frac{9}{4}$. (ii) $x = \pm 2\sqrt{2}$.

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Completed-square Form and Parabola Features

WHAT THIS FAMILY ASKS

Convert a quadratic to completed-square form, read its turning point and symmetry, and use these features in sketches, intersections or region descriptions.

VARIANT ROUTE

V01 Complete the Square, Read and Sketch

V02 Complete the Square, then Intersect and Define a Region

5 Complete Classified questions

39 total marks

2.

$$f(x) = 11 - 4x - 2x^2$$

- (a) Express $f(x)$ in the form

$$a + b(x + c)^2$$

where a , b and c are integers to be found.

(3)

- (b) Sketch the graph of the curve C with equation $y = f(x)$, showing clearly the coordinates of the point where the curve crosses the y -axis.

(2)

- (c) Write down the equation of the line of symmetry of C .

(1)

(Total 6 marks)

Question 4

WMA11/01 January 2022

T02 Quadratics

F02 Completed-square Form
and Parabola FeaturesV01 Complete the Square,
Read and Sketch

1

Complete the square

$$f(x) = 11 - 4x - 2x^2 = -2(x^2 + 2x) + 11.$$

2

Write in vertex form

$$-2(x^2 + 2x) + 11 = -2((x + 1)^2 - 1) + 11 = 13 - 2(x + 1)^2.$$

3

Identify the graph features

The graph is a downward-opening parabola with maximum point $(-1, 13)$.
The y-intercept is $f(0) = 11$, so it crosses the y-axis at $(0, 11)$.

4

State the line of symmetry

From $13 - 2(x + 1)^2$, the line of symmetry is $x = -1$.

FINAL ANSWER

(a) $13 - 2(x + 1)^2$. (b) **y-intercept** $(0, 11)$. (c) $x = -1$.

3. (a) Express $3x^2 + 12x + 13$ in the form

$$a(x + b)^2 + c$$

where a , b and c are integers to be found.

(3)

- (b) Hence sketch the curve with equation $y = 3x^2 + 12x + 13$

On your sketch show clearly

- the coordinates of the y intercept
- the coordinates of the turning point of the curve

(3)

(Total for Question 3 is 6 marks)

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Question 5

WMA11/01 May/June 2023

T02 Quadratics

F02 Completed-square Form
and Parabola FeaturesV01 Complete the Square,
Read and Sketch

1

Complete the square

$$3x^2 + 12x + 13 = 3(x^2 + 4x) + 13.$$

2

Use $(x + 2)^2$

$$x^2 + 4x = (x + 2)^2 - 4, \text{ so } 3(x^2 + 4x) + 13 = 3(x + 2)^2 - 12 + 13.$$

3

Read the features

$y = 3(x + 2)^2 + 1$ has turning point $(-2, 1)$. The y-intercept is found from $x = 0$, giving $y = 13$.

4

Sketch description

The curve is an upward-opening parabola with minimum $(-2, 1)$ and y-intercept $(0, 13)$.

FINAL ANSWER

(a) $3(x + 2)^2 + 1$. (b) **y-intercept** $(0, 13)$ and **turning point** $(-2, 1)$.

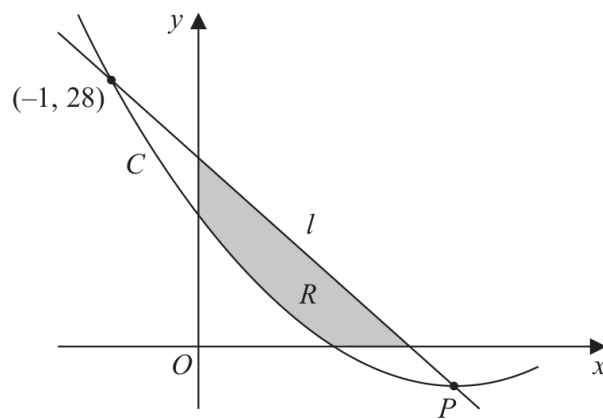


Figure 5

Figure 5 shows part of the curve C with equation $y = f(x)$ where

$$f(x) = 2x^2 - 12x + 14$$

- (a) Write $2x^2 - 12x + 14$ in the form

$$a(x + b)^2 + c$$

where a , b and c are constants to be found.

(3)

Given that C has a minimum at the point P

- (b) state the coordinates of P

(1)

The line l intersects C at $(-1, 28)$ and at P as shown in Figure 5.

- (c) Find the equation of l giving your answer in the form $y = mx + c$ where m and c are constants to be found.

(3)

The finite region R , shown shaded in Figure 5, is bounded by the x -axis, l , the y -axis, and C .

- (d) Use inequalities to define the region R .

(3)

(Total for Question 11 is 10 marks)

Question 6

WMA11/01 October 2023

T02 Quadratics

F02 Completed-square Form
and Parabola FeaturesV02 Complete the Square,
then Intersect and Define a
Region

1

Complete the square

$$2x^2 - 12x + 14 = 2(x^2 - 6x) + 14 = 2((x - 3)^2 - 9) + 14 = 2(x - 3)^2 - 4.$$

2

State the minimum point

The minimum of $2(x - 3)^2 - 4$ is at $P = (3, -4)$.

3

Find the gradient of l

The line passes through $(-1, 28)$ and $P(3, -4)$, so $m = \frac{-4 - 28}{3 - (-1)} = \frac{-32}{4} = -8$.

4

Find the equation of l

Using $P(3, -4)$, $y + 4 = -8(x - 3)$, so $y = -8x + 20$.

5

Use the axes as boundaries

The region is to the right of the y-axis and above the x-axis, so $x \geq 0$ and $y \geq 0$.

6

Use the line and curve as boundaries

The region is below the line and above the curve, so $y \leq -8x + 20$ and $y \geq 2x^2 - 12x + 14$.

FINAL ANSWER

(a) $2(x - 3)^2 - 4$. (b) $P = (3, -4)$. (c) $y = -8x + 20$. (d) $x \geq 0$, $y \geq 0$, $y \leq -8x + 20$, $y \geq 2x^2 - 12x + 14$.

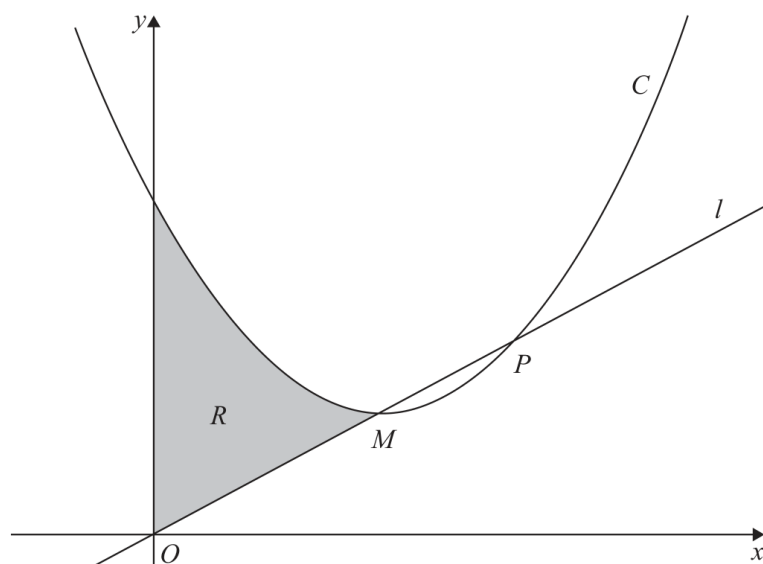


Figure 1

**In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

- (a) Express $2x^2 - 16x + 50$ in the form

$$a(x + b)^2 + c$$

where a , b and c are constants to be found.

(3)

Figure 1 shows a sketch of the curve C with equation $y = 2x^2 - 16x + 50$

Given that in Figure 1

- M is the minimum point on C
- line l passes through the origin and intersects C at the points M and P

- (b) find, using algebra and showing your working, the x coordinate of P .

(5)

The region R is shown shaded in Figure 1.

- (c) Use inequalities to fully define R .

(2)

(Total for Question 5 is 10 marks)

Question 7

WMA11/01 October 2025

T02 Quadratics

F02 Completed-square Form
and Parabola FeaturesV02 Complete the Square,
then Intersect and Define a
Region

1

Complete the square

$$2x^2 - 16x + 50 = 2(x^2 - 8x) + 50 = 2[(x - 4)^2 - 16] + 50.$$

2

Simplify the completed square

$$2x^2 - 16x + 50 = 2(x - 4)^2 + 18.$$

3

Find MThe minimum point is $M = (4, 18)$.

4

Find the line through O and M

Since l passes through $(0, 0)$ and $(4, 18)$, its gradient is $\frac{18}{4} = \frac{9}{2}$, so $l : y = \frac{9}{2}x$.

5

Find the second intersection

Set $2x^2 - 16x + 50 = \frac{9}{2}x$.

6

Form the quadratic

Multiplying by 2 gives $4x^2 - 32x + 100 = 9x$, so $4x^2 - 41x + 100 = 0$.

7

Factor and select P

$4x^2 - 41x + 100 = (x - 4)(4x - 25) = 0$. The point M has $x = 4$, so P has $x = \frac{25}{4}$.

8

Define the region

The shaded region is above the line, below the curve, and between the y-axis and M .

9

Write inequalities

$\frac{9}{2}x \leq y \leq 2x^2 - 16x + 50$ with $0 \leq x \leq 4$.

FINAL ANSWER

(a) $2(x-4)^2 + 18$. (b) $x_P = \frac{25}{4}$. (c) $\frac{9}{2}x \leq y \leq 2x^2 - 16x + 50, 0 \leq x \leq 4$.

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5. The curve C has equation

$$y = -2x^2 + 12x + 7$$

- (a) Write $-2x^2 + 12x + 7$ in the form $a(x + b)^2 + c$, where a , b and c are constants. (3)

The point P is the maximum point on C .

- (b) Find the coordinates of P . (2)

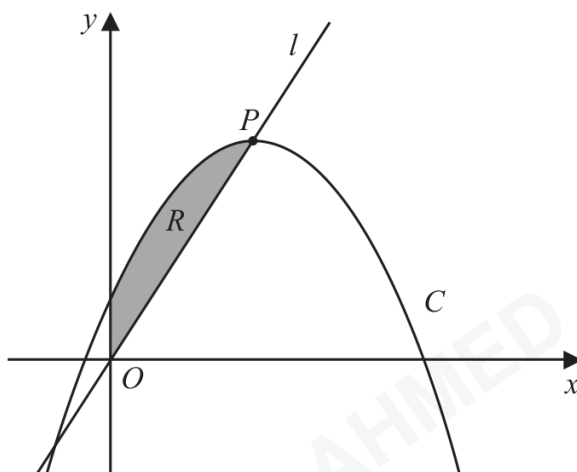


Figure 1

Figure 1 shows a sketch of the curve C and a line l .

The line l passes through the point P and the origin O .

The region R , shown shaded in Figure 1, is bounded by C , l and the y -axis.

- (c) Use inequalities to define R . (2)

(Total for Question 5 is 7 marks)

Question 8

WMA11/01 May/June 2026

T02 Quadratics

F02 Completed-square Form
and Parabola FeaturesV02 Complete the Square,
then Intersect and Define a
Region

1

(a) Factor out the leading coefficient

$$-2x^2 + 12x + 7 = -2(x^2 - 6x) + 7.$$

2

(a) Complete the square

$$x^2 - 6x = (x - 3)^2 - 9, \text{ so } -2x^2 + 12x + 7 = -2((x - 3)^2 - 9) + 7 = -2(x - 3)^2 + 25.$$

3

(b) Read the maximum from completed-square form

Because the coefficient of $(x - 3)^2$ is negative, the curve has its maximum when $(x - 3)^2 = 0$. Hence $x = 3$, and the maximum value is $y = 25$, so $P = (3, 25)$.

4

(c) Find the equation of the boundary line

The line through $O = (0, 0)$ and $P = (3, 25)$ has gradient $\frac{25}{3}$, so l has equation $y = \frac{25}{3}x$.

5

(c) Describe the shaded region

Points in R are below the curve, above the line, and on or to the right of the y -axis. Therefore $y \leq -2x^2 + 12x + 7$, $y \geq \frac{25}{3}x$, and $x \geq 0$.

FINAL ANSWER

$$(a) -2(x - 3)^2 + 25. \quad (b) P = (3, 25). \quad (c) y \leq -2x^2 + 12x + 7, y \geq \frac{25}{3}x, x \geq 0.$$

Construct a Quadratic from Graph Data

WHAT THIS FAMILY ASKS

Select a root form or vertex form from graph information and determine its scale using another point or turning value.

VARIANT ROUTE

V01 Roots, Symmetry and a Point

V02 Vertex Form and a Point

2 Complete Classified questions

18 total marks

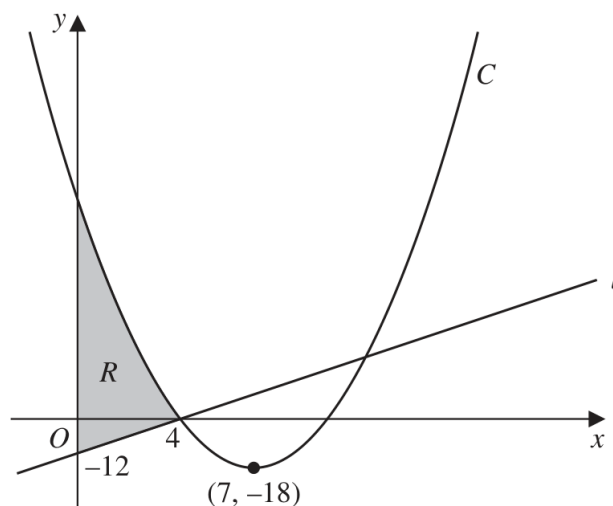


Figure 2

Figure 2 shows a sketch of the straight line l and the curve C .

Given that l cuts the y -axis at -12 and cuts the x -axis at 4 , as shown in Figure 2,

- (a) find an equation for l , writing your answer in the form $y = mx + c$, where m and c are constants to be found.

(2)

Given that C

- has equation $y = f(x)$ where $f(x)$ is a quadratic expression
- has a minimum point at $(7, -18)$
- cuts the x -axis at 4 and at k , where k is a constant

- (b) deduce the value of k ,

(1)

- (c) find $f(x)$.

(3)

The region R is shown shaded in Figure 2.

- (d) Use inequalities to define R .

(2)

(Total for Question 8 is 8 marks)

Question 9

WMA11/01 January 2023

T02 Quadratics

F03 Construct a Quadratic
from Graph DataV01 Roots, Symmetry and a
Point

1

Find the straight line

The line cuts the y-axis at -12 and the x-axis at 4 , so its gradient is $\frac{0 - (-12)}{4 - 0} = 3$. Hence l is $y = 3x - 12$.

2

Use symmetry of the quadratic

The minimum point has x-coordinate 7 . The roots are 4 and k , so their mid-point is 7 .

3

Find k

$\frac{4 + k}{2} = 7$, so $k = 10$.

4

Build the quadratic

Let $f(x) = a(x - 4)(x - 10)$. Since the minimum point is $(7, -18)$, $-18 = a(3)(-3) = -9a$, so $a = 2$.

5

Write $f(x)$

$f(x) = 2(x - 4)(x - 10) = 2x^2 - 28x + 80$.

6

Identify the shaded region

The region lies between the y-axis and $x = 4$, above the line and below the curve: $0 \leq x \leq 4$ and $3x - 12 \leq y \leq 2x^2 - 28x + 80$.

FINAL ANSWER

(a) $y = 3x - 12$. (b) $k = 10$. (c) $f(x) = 2(x - 4)(x - 10)$. (d) $0 \leq x \leq 4$, $3x - 12 \leq y \leq 2x^2 - 28x + 80$.

9. The curve C_1 has equation $y = f(x)$.

Given that

- $f(x)$ is a quadratic expression
- C_1 has a maximum turning point at $(2, 20)$
- C_1 passes through the origin

(a) sketch a graph of C_1 showing the coordinates of any points where C_1 cuts the coordinate axes, (2)

(b) find an expression for $f(x)$. (3)

The curve C_2 has equation $y = x(x^2 - 4)$

Curve C_1 and C_2 meet at the origin, and at the points P and Q

Given that the x coordinate of the point P is negative,

(c) using algebra and showing all stages of your working, find the coordinates of P (5)

(Total for Question 9 is 10 marks)

Question 10

WMA11/01 January 2024

T02 Quadratics

F03 Construct a Quadratic from Graph Data
V02 Vertex Form and a Point

1

Use symmetry for the sketch

The maximum is at $(2, 20)$ and the curve passes through $(0, 0)$. Since the axis of symmetry is $x = 2$, the other x-intercept is $(4, 0)$.

2

Use completed-square form

Let $f(x) = a(x - 2)^2 + 20$.

3

Use the origin

$0 = a(0 - 2)^2 + 20$, so $4a = -20$ and $a = -5$.

4

Write $f(x)$

$$f(x) = -5(x - 2)^2 + 20 = -5x^2 + 20x = -5x(x - 4).$$

5

Equate the two curves

$$-5x^2 + 20x = x(x^2 - 4) = x^3 - 4x.$$

6

Factorise

$$x^3 + 5x^2 - 24x = 0, \text{ so } x(x^2 + 5x - 24) = 0 \text{ and } x(x + 8)(x - 3) = 0.$$

7

Use the negative x-coordinate

The negative x-coordinate is $x = -8$.

8

Find y

$$y = x(x^2 - 4) = (-8)(64 - 4) = -480, \text{ so } P = (-8, -480).$$

FINAL ANSWER

(a) **downward parabola through $(0, 0)$ and $(4, 0)$ with maximum $(2, 20)$.** (b) $f(x) = -5x(x - 4)$. (c) $P = (-8, -480)$.

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Discriminant and Root-count Conditions

WHAT THIS FAMILY ASKS

Translate a required number of real roots or curve intersections into a sign condition on the discriminant, then solve the resulting parameter equation or inequality.

VARIANT ROUTE

V01 Parameter Range from a Required Root Count

V02 Curve Tangency or Separation

4 Complete Classified questions

23 total marks

4. Given that the equation

$$kx^2 + 6kx + 5 = 0 \quad \text{where } k \text{ is a non zero constant}$$

has no real roots, find the range of possible values for k .

(4)

(Total for Question 4 is 4 marks)

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Question 11

WMA11/01 January 2023

T02 Quadratics

F04 Discriminant and Root-count Conditions

V01 Parameter Range from a Required Root Count

1

Use the discriminant conditionFor no real roots, the discriminant must be negative: $b^2 - 4ac < 0$.

2

Substitute a, b, and cFor $kx^2 + 6kx + 5 = 0$, $a = k$, $b = 6k$, and $c = 5$.

3

Form the inequality $(6k)^2 - 4(k)(5) < 0$, so $36k^2 - 20k < 0$.

4

Solve the inequality $4k(9k - 5) < 0$, so the product is negative when $0 < k < \frac{5}{9}$.**FINAL ANSWER**

$$0 < k < \frac{5}{9}.$$

6.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

The equation

$$4(p - 2x) = \frac{12 + 15p}{x + p} \quad x \neq -p$$

where p is a constant, has two distinct real roots.

(a) Show that

$$3p^2 - 10p - 8 > 0 \quad (3)$$

(b) Hence, using algebra, find the range of possible values of p (3)

(Total for Question 6 is 6 marks)

Question 12

WMA11/01 October 2023

T02 Quadratics

F04 Discriminant and Root-count Conditions

V01 Parameter Range from a Required Root Count

1

Clear the denominator

$$4(p - 2x) = \frac{12 + 15p}{x + p} \text{ gives } 4(p - 2x)(x + p) = 12 + 15p.$$

2

Form a quadratic in x

$$\text{Expanding gives } 4p^2 - 4px - 8x^2 = 12 + 15p, \text{ so } 8x^2 + 4px + 12 + 15p - 4p^2 = 0.$$

3

Use the discriminant condition

$$\text{For two distinct real roots, } b^2 - 4ac > 0.$$

4

Simplify the inequality

$$(4p)^2 - 4(8)(12 + 15p - 4p^2) > 0, \text{ so } 144p^2 - 480p - 384 > 0.$$

5

Divide and factorise

$$\text{Dividing by 48 gives } 3p^2 - 10p - 8 > 0, \text{ and } 3p^2 - 10p - 8 = (3p + 2)(p - 4).$$

6

Solve the discriminant inequality

The upward-opening quadratic is positive outside its roots, so the discriminant condition gives $p < -\frac{2}{3}$ or $p > 4$.

7

Restore the denominator restriction

Cross-multiplication is only equivalent when $x \neq -p$. Substituting the forbidden value $x = -p$ into the quadratic gives $12 + 15p$, so $x = -p$ becomes an extraneous quadratic root when $p = -\frac{4}{5}$. At that value the original equation has only one valid root, so it must be excluded.

8

State the corrected range

Therefore the exact domain-valid range is $p < -\frac{2}{3}$ with $p \neq -\frac{4}{5}$, or $p > 4$.

Equivalently, $p \in (-\infty, -\frac{4}{5}) \cup (-\frac{4}{5}, -\frac{2}{3}) \cup (4, \infty)$.

FINAL ANSWER

$$p < -\frac{2}{3}, p \neq -\frac{4}{5}, \text{ or } p > 4.$$

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10.

**In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

$$(k-1)x^6 + 4x^3 + (k-4) = 0 \quad \text{where } k \text{ is a constant}$$

- (a) Find the exact solutions to the given equation for $k = 4.5$ (3)
- (b) Find the set of possible values of k for which the given equation has no real roots. (4)

(Total for Question 10 is 7 marks)

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Question 13

WMA11/01 May/June 2025

T02 Quadratics

F04 Discriminant and Root-count Conditions

V01 Parameter Range from a Required Root Count

1

Substitute k equals 4.5The equation becomes $3.5x^6 + 4x^3 + 0.5 = 0$.

2

Clear decimalsMultiplying by 2 gives $7x^6 + 8x^3 + 1 = 0$.

3

Use a quadratic in x cubed $7x^6 + 8x^3 + 1 = (x^3 + 1)(7x^3 + 1) = 0$.

4

Solve for x $x^3 = -1$ or $x^3 = -\frac{1}{7}$, so $x = -1$ or $x = -\sqrt[3]{\frac{1}{7}}$.

5

View the equation as a quadraticLet $u = x^3$. Then the equation is $(k - 1)u^2 + 4u + (k - 4) = 0$.

6

Use no real rootsSince every real value of u gives a real x , the original equation has no real roots exactly when this quadratic in u has no real roots.

7

Use the discriminantRequire $b^2 - 4ac < 0$: $16 - 4(k - 1)(k - 4) < 0$.

8

Simplify

$16 - 4(k^2 - 5k + 4) < 0$ **gives** $20k - 4k^2 < 0$, **so** $4k(5 - k) < 0$.

9

Solve the inequality

$4k(5 - k) < 0$ **for** $k < 0$ **or** $k > 5$.

FINAL ANSWER

(a) $x = -1, -\sqrt[3]{\frac{1}{7}}$. (b) $k < 0$ **or** $k > 5$.

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4. The curve C_1 has equation

$$y = x^2 + kx - 9$$

and the curve C_2 has equation

$$y = -3x^2 - 5x + k$$

where k is a constant.

Given that C_1 and C_2 meet at a single point P

(a) show that

$$k^2 + 26k + 169 = 0 \quad (3)$$

(b) Hence find the coordinates of P (3)

(Total for Question 4 is 6 marks)

Question 14

WMA11/01 May/June 2024

T02 Quadratics

F04 Discriminant and Root-
count ConditionsV02 Curve Tangency or Sepa-
ration

1

Equate the curves

$$x^2 + kx - 9 = -3x^2 - 5x + k \text{ gives } 4x^2 + (k + 5)x - (k + 9) = 0.$$

2

Use one intersection only

For a single point of intersection, the discriminant is zero.

3

Form the condition on k

$$(k + 5)^2 - 4(4)(-(k + 9)) = 0, \text{ so } k^2 + 26k + 169 = 0.$$

4

Find k

$$k^2 + 26k + 169 = (k + 13)^2, \text{ so } k = -13.$$

5

Find x

$$\text{Substituting } k = -13 \text{ gives } 4x^2 - 8x + 4 = 0, \text{ so } 4(x - 1)^2 = 0 \text{ and } x = 1.$$

6

Find y

$$y = x^2 + kx - 9 = 1 - 13 - 9 = -21, \text{ so } P = (1, -21).$$

FINAL ANSWER

$$P = (1, -21).$$

Quadratic Modelling from Constraints

WHAT THIS FAMILY ASKS

Translate linked real-world constraints into a quadratic equation and select the solution satisfying the original meaning.

VARIANT ROUTE

V01 Sum-product Geometric Constraints

1 Complete Classified questions

6 total marks

2.

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

A rectangular sports pitch has length x metres and width y metres, where $x > y$

Given that the perimeter of the pitch is 350 m,

(a) write down an equation linking x and y (1)

Given also that the area of the pitch is 7350 m^2

(b) write down a second equation linking x and y (1)

(c) hence find the value of x and the value of y (4)

(Total for Question 2 is 6 marks)

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Question 15

WMA11/01 May/June 2023

T02 Quadratics

F05 Quadratic Modelling from Constraints

V01 Sum-product Geometric Constraints

1

Use the perimeter

$$2x + 2y = 350, \text{ so } x + y = 175.$$

2

Use the area

$$xy = 7350.$$

3

Form a quadratic

$$\text{Using } y = 175 - x, x(175 - x) = 7350, \text{ so } x^2 - 175x + 7350 = 0.$$

4

Solve

$$x = \frac{175 \pm \sqrt{175^2 - 4(7350)}}{2} = \frac{175 \pm 35}{2}, \text{ giving } x = 105 \text{ or } 70.$$

5

Use $x > y$

$$\text{Since } x > y, x = 105 \text{ and } y = 70.$$

FINAL ANSWER

$$x = 105, y = 70.$$